Evaluating Future Paths of Debt in the United States

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1 Developing a Model of Debt

In our previous paper on the sustainability of U.S. public debt, we showed that conditions formerly favorable to sustainability, most notably $r < g$ (the interest rate on the debt lower than the economy's growth path), have turned less favorable in recent years. We also argued that debt sustainability and the trajectory of the debt itself had been worsened by Congressional actions to deficit-finance historically large tax cuts and spending, including the recently passed One Big Beautiful Bill Act (OBBBA), which according to the Yale Budget Lab, will raise the debt-to-GDP ratio by 52 percentage points from its baseline 30 years from now.

One challenge in the debt sustainability literature is that there are no widely accepted metrics to distinguish the sustainability of one budget path relative to another. Historically large increases in debt-to-GDP ratios are certainly a signal of sustainability problems, but many countries, most notably Japan, have experienced exactly such increases in their debt-to-GDP ratio without much pressure on interest rates or other obviously distortionary impacts.

We introduced a new warning criterion for unsustainability: "ahistorical adjustments" (AA), meaning a situation wherein external factors, such as a sharp rise in term premia on government debt, necessitated a sharp, quick adjustment to the fiscal path, say by reducing primary deficits. When that adjustment was "large" in historical terms — i.e., in the right tail of the empirical distribution of historical adjustments — we argued that the path was potentially unsustainable. Intuitively, if the debt ratio or real net interest as a share of GDP — another tracking metric we employ — strays far above its historical path, a sharp adjustment is more likely to be painful economically and challenging politically.

In this paper, in order to more rigorously identify the factors that determine debt sustainability and what constitutes an AA, we devise and run a simulation model based on the basic debt determinants that generates a distribution of fiscal outcomes. The model's parameters enable us to simulate Congress's level of concern about the fiscal outlook, the 'gravity' of the neutral interest rate r^*, the persistence of interest rates over time, and — most importantly — alternative assumptions about the mean future path of growth, interest rates, and the primary deficit. For example, in Subsection 1.6, we show how our model can be used to project forward; specifically, we compare the implications of using either the pre-One Big Beautiful Bill Act (OBBBA) CBO deficit projections or the post-OBBBA Yale Budget Lab deficit projections in our model.

We find that under the vast majority of our simulations an AA will be needed to achieve sustainability. For example, [I'll explain]

The model we use is a variation of a class of probabilistic models often referred to as "Stochastic Debt Sustainability Analysis" (SDSA), as in Blanchard (2024) and Favero (2025). We adopt a straightforward set of equations that simulates paths for the relevant debt determinants: r, g, and s (interest rate, growth rate, primary balance, respectively).[1] Our goal is to build and automate a model that can be useful to fiscal policymakers considering measures that would change the fiscal path. Ours is a similar approach to the IMF's Sovereign Risk and Debt Sustainability Analysis for Market Access Countries and the European Commission's Debt Sustainability Monitor.

The following system of equations allows us — with assumptions including exogenous forecasts — to run Monte Carlo simulations that result in a distribution of future debt paths. We spend the rest of this section explaining these equations and their logic.

$$b_t = b_{t-1} + \frac{r_t^{av} - g_t}{1 + g_t} b_{t-1} - s_t \tag{1}$$

$$s_t = (1 - c)a_s + c(r_t^{av} - g_t)b_{t-1} + e_t^s, \quad e_t^s \sim \mathcal{N}(0, s_s) \tag{2}$$

$$g_t = a_{gt} + x_{gt} + e_t^g, \quad e_t^g \sim \mathcal{N}(0, s_g) \tag{3}$$

$$x_{gt} = x_{t-1} + e_t^x, \quad e_t^x \sim \mathcal{N}(0, s_x) \tag{4}$$

$$r_t = r^* + \beta_r b_{t-1} + \rho(r_{t-1} - r^* - \beta_r b_{t-2}) + e_t^r, \quad e_t^r \sim \mathcal{N}(0, s_r) \tag{5}$$

$$\Delta r_t = \beta_r(\Delta b_{t-1} - \rho \Delta b_{t-2}) + \rho \Delta r_{t-1} + (\epsilon_t - \epsilon_{t-1}) \tag{6}$$

$$r_t^{av} = \sigma \cdot r_{t-1}^{av} + (1 - \sigma) \cdot r_t \tag{7}$$

$$\tag{8}$$

1.1 Our Approach to Modeling the Interest Rate r

Here we present a simplified explanation for our modeling of r; for a more detailed explanation, see Section 3.2 in the Appendix. We model interest rates' *first difference* (thus avoiding the confounding factors of the secular trends in interest rates and debt) as a function of the long-run equilibrium real rate r^*. From there, we estimate the average interest rate paid on all outstanding debt, r_t^{av}, as a function of the previous period's average interest rate r_{t-1}^{av} and a 'stickiness' parameter σ that governs how contemporaneous current interest rates are.[2]

This setup requires the initial calibration of three parameters: r^*, ρ, and β_r. r^*, the real neutral interest rate, is not directly observable but has been estimated by Holston, Laubach, and Williams, who find a post-Great Financial Crisis (GFC) average r^* of 0.98%, which we round to 1% in our model (Holston, Laubach, and Williams 2017).[3] Note that this is consistent with the Federal Reserve's most recent Survey of Economic Projections which has nominal r* at 3% (1% real plus the Fed's 2% inflation target). However, as there are reasons to believe that r^* may have climbed in recent years (see Carvalho and Nechio 2025 and Benigno and Sandri 2024), we simulate models with higher initial calibrations.

Second, we have to determine ρ, which governs the persistence of how the interest rate deviates from its neutral value, r^*, and β_r, which governs the magnitude of the pass-through effect of changes in debt on changes in the interest rate. Our estimation strategy for these parameters, relying on first-differencing to

[1] See Celasun and Ostry 2006; Cottarelli and Moghadam 2011; Berti 2013; Demirel and Otterson 2022; and Auerbach and Yagan 2025. Celasun and Ostry 2006, employing a "fan chart" approach, were particularly pioneering for introducing a probabilistic approach to the then-standard debt sustainability framework of "deterministic bound-testing". The primary motivation for opting for a probabilistic approach is that we don't rely on a few stylized exogenous shocks to generate a range of outcomes, allowing for greater flexibility and more fidelity.

[2] We fix $\sigma = 0.8$. In practice, σ will vary in accordance with the maturity profile of outstanding debt. We do not account for such dynamics in our model.

[3] See Figure 5 for the time-series of the HLW-estimated values of r^*.

avoid our model being overcome by the secular downwards trend in the term premium and upwards trend in the debt level, mirrors that of Gust and Skaperdas 2024. We estimate the following equation:

$$\Delta y_t = \beta_r \Delta b_{t-1} - \beta_r \rho \Delta b_{t-2} + \rho \Delta y_{t-1} + \gamma' \Delta \mathbf{X_t} + \epsilon_t \tag{9}$$

Where y_t is the 10-year ACM term premium (Adrian and Moench 2013), b_{t-1} the lagged debt/GDP ratio, and $\mathbf{X_t}$ a vector of controls (we keep this lean to avoid overfitting in our model, so that only output gap and primary deficit / GDP, s_t, are included).[4] Then, we set ρ to be equal to the coefficient on Δy_{t-1}.[5] The full results from estimating Equation 9 can be found in Table 4. The results, which appear to be robust to cutting the sample up around the GFC, show that $\hat{\beta}_r = 0.017$ — meaning a one percentage point quarterly increase in the debt as a percentage of GDP will translate to a 1.7 basis point increase in the term premium — and $\hat{\rho} = 0.158$. This estimate $\hat{\beta}_r$ falls within the historic estimates for the parameter (see 3), although certainly in the left side of the distribution of previous estimates. This suggests that, in effect, our debt projects should be interpreted as lower bounds; a higher value for $\hat{\beta}_r$ would only amplify any debt path. We accordingly use these values when running our SDSA Monte Carlo simulations.

1.2 Our Approach to Modeling the Primary Surplus/Deficit s

Here we present a simplified explanation for our modeling of s; for a more detailed explanation, see Section 3.3 in the Appendix. We model primary deficits as a function of a Congressional policy responsiveness parameter c, the mean forecasted primary deficit a_s, a noise term e^s_t, and, in cases where the policy responsiveness parameter $c \neq 0$, the gap between the average interest rate paid on outstanding debt and growth $(r^{av}_t - g_t)$. To reflect the fiscal impacts of the recently-passed One Big Beautiful Bill Act (OBBBA), we adopt The Budget Lab's primary deficit projections conditioned on the permanent enactment of the Senate version of the bill as the average of future distributions of the primary deficit.[6]

One of the key parameters in the models is c, the policy responsiveness parameter. Put simply, this parameter dictates the weight given by policymakers to taking steps to stabilize the debt (so that $b - b_{-1} \sim 0$), absent fluctuations because of noise terms) in determining the primary surplus. Auerbach and Yagan (2025) find that, whereas between 1984-2003 there is evidence of Congress reducing the deficit in reaction to rising projected deficits, any such feedback effect disappeared from 2004-2024 (Auerbach and Yagan 2025). In particular, they find that from 1984-2003, Congress reacted to a 1% of GDP higher CBO-projected deficit with 0.15% of GDP deficit reduction. Then, from 2004-2024, a 1% of GDP higher CBO-projected deficit was associated with deficit reduction equal to -0.03% (in practice, a null effect).

This result suggests that in its current state, Congress's fiscal policymaking is entirely inelastic with respect to the debt level ($c = 0$). Recent qualitative evidence would support this view. To illustrate the need for fiscal moderation, we run our model under three assumptions of c's true value. In the first, $c = 0$ and we assume that, as Auerbach and Yagan (2025) found, Congress's fiscal policymaking is entirely unresponsive to the level of debt for the next 10 years. This we refer to as an 'irresponsible' fiscal regime hereafter. Then, as one counterfactual, we model a more 'responsible' fiscal regime, which — effective immediately — returns to the pre-2003 Congressional behavior as estimated by Auerbach and Yagan. For this model of a 'responsible' regime, we let $c = 0.15$; this *approximates* 15% of deficit-setting being accounted for by attempts to stabilize the debt. Finally, we run our model with $c = 0.3$, meant to represent what we perceive of as an upper bound of possible future levels of fiscal responsibility. What we find is that even a return to the fiscal posture of 1984-2003 (and beyond, the elevation of fiscal responsibility to its upper bound) would be inadequate to sufficiently address debt sustainability and avoid the need for an ahistorical adjustment.

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[4] The term premium is in percentage points, debt is expressed as a percent of GDP, and all Δ values are quarter-to-quarter changes in percentage point. Thus β_r should be interpreted as the effect of a one percentage point quarterly change in the debt as a percentage of GDP on the term premium.

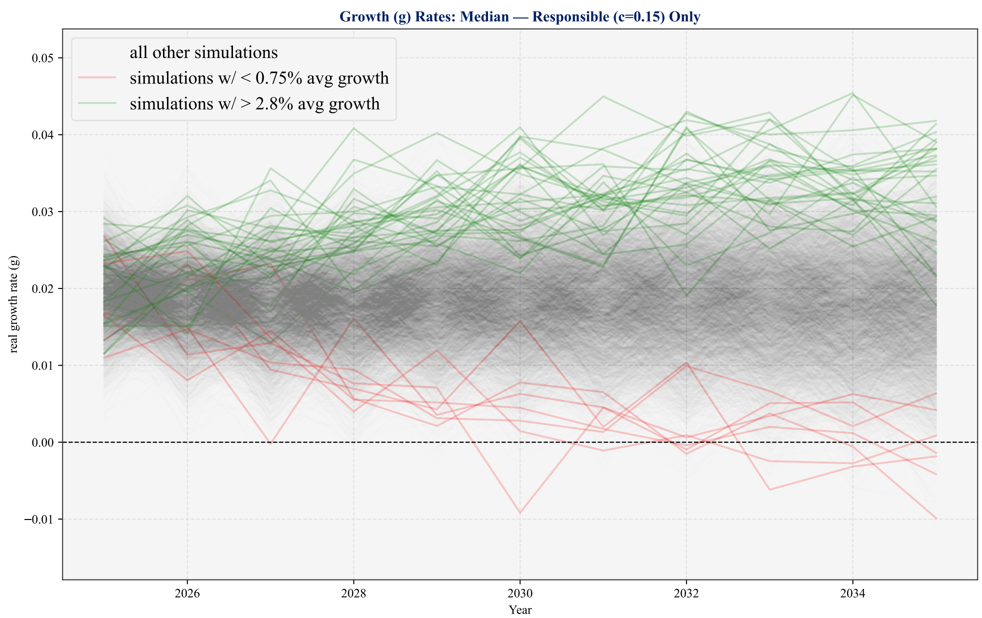
[5] Specifically, we estimate Equation 9 via Ordinary-Least-Squares (OLS) with Newey-West HAC standard errors.

[6] This approach to modeling debt policy takes a fundamentally different approach from models whose fiscal policy is incorporated into a *reaction function* framework (e.g., Celasun and Ostry 2006). Rather than calibrate a model estimated using a country panel dataset, we believe the U.S.'s fiscal situation is unique enough to warrant its own model. There is little historical evidence of a measureable fiscal reaction process — perhaps because, unlike the emerging markets often studied by economists at the IMF, for example, the U.S. is not typically or observably punished by capital markets for excessive borrowing. Instead, we leverage existing forecasts as the center of a distribution whose shape is second moments (shape) are determined by probabilistic shocks.

1.3 Our Approach to Modeling the Real Growth Rate g

We model growth so that it is composed of three parts. First, the first moment (mean) of growth's distribution in year t is exogenously defined by the vector a_{gt}. We set a_{gt} equal to the Congressional Budget Office's (CBO) baseline forecast of real GDP growth. Second, the growth in any period can be shifted by a *transitory* shock, e_t^g, randomly drawn from a normal distribution. Third, the path of growth can be shifted by a *persistent* shock, x_{gt}, which moves the growth rate of the economy onto a different path. This results in a range of growth outcomes, including the most favorable (where shocks are consistently positive) and the least favorable (where shocks are consistently negative).

To illustrate how this works in practice, below I've plotted the full range (across 5,000 simulations) of growth paths that our model produces. I've highlighted the high-growth paths (where average growth is higher than 2.8% over the next 10 years) and the low-growth paths (where average growth is lower than 0.75% over the next 10 years). As we observe, these growth paths compound over time, as repeated positive (or negative) shocks push the path further and further from the CBO baseline rate.



1.4 Results

Running the model with 5,000 simulations illustrates persistently positive trends in debt accumulation. First, in Figure 1, we've plotted the level of debt as a % of GDP (with the interquartile range in every year shaded in corresponding color to the median, represented by the singular bold line). In this case, we vary the c parameter as described above to match 'irresponsible' ($c = 0$), 'responsible' ($c = 0.15$), and 'very responsible' ($c = 0.3$) fiscal policymaking regimes.

Next, in Figure 2 we plot the change in the slopes (the **acceleration**) of the three scenarios in Figure 1. In each case, the median fiscal path's debt ratio accelerates following five years, a sign of unsustainability (though some simulations below the 50th percentile decelerate).

These results reveal that even were Congress to *suddenly* re-discover the fiscal rectitude of the 1984-2003 period (as per Auerbach and Yagan), debt would still relentlessly increase. A positive *and increasing* curvature to the debt path (as shown in Figure 2) suggests a spiral could occur; while not an exact estimate of the cost of debt servicing (a true estimate of the cost of debt servicing would require an enrichment of the model to incorporate the changing composition of outstanding debt), this curvature plot suggests that under plausible interest rate assumptions, debt servicing costs would consistently rise as a share of the economy in the vast majority of simulated futures.

Recall our emphasis on ahistorical adjustments, meaning debt reduction of a historically large magnitude. To illustrate this concept using our simulated results, we plot in Figure 3 the distribution of 10-year changes in the debt-to-GDP ratio (in terms of percentage points) for 1) historical data (1970 - present), 2) our irresponsible fiscal regime simulations, and 3) our responsible fiscal regime solutions.

These simulated distributions show that the medians of the $c = 0$ and $c = 0.15$ cases to be at the very far end of the historical distribution. The highly responsible regime ($c = 0.3$) does better, with the left tail not too far above the historical median. What this says in practical terms is that to avoid an AA should we hit a debt crisis, Congress would be have to be far more responsible (in fact, $2\times$) than they've been in any decades, and very lucky (i.e., consistently get positive shocks to growth and rates and find themselves in the left tail of the distribution of debt-to-GDP ratios). Our simulations thus suggest we are heading for uncharted waters.

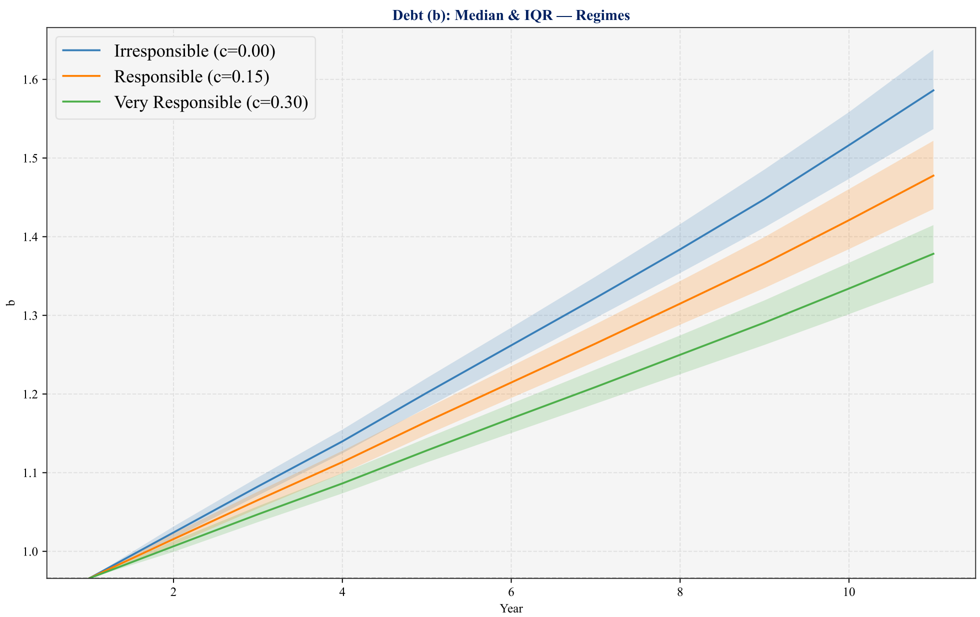


Figure 1: Debt Level Projections (as % of GDP)

Below I present two alternative ways of comparing the distribution of these modelled projections and the observed historical distributions:

Table 1: Empirical percentiles (10th–90th) of 10-year debt/GDP changes: historical and simulated distributions.

Distribution	10th	25th	50th	75th	90th
$c = 0.00$	52.7	57.1	62.0	67.2	71.8
$c = 0.15$	43.3	46.9	51.2	55.6	59.5
$c = 0.30$	34.5	37.6	41.3	44.9	48.1
Historical	-11.8	1.9	13.8	25.0	37.5

Table 2: Percentile ranks of simulated 10-year debt/GDP changes within the historical distribution.

Simulation	10th Percentile Rank in Hist. (%)	Median Rank in Hist. (%)	90th Percentile Rank in Hist. (%)
$c = 0.00$	100.0	100.0	100.0
$c = 0.15$	100.0	100.0	100.0
$c = 0.30$	84.4	100.0	100.0

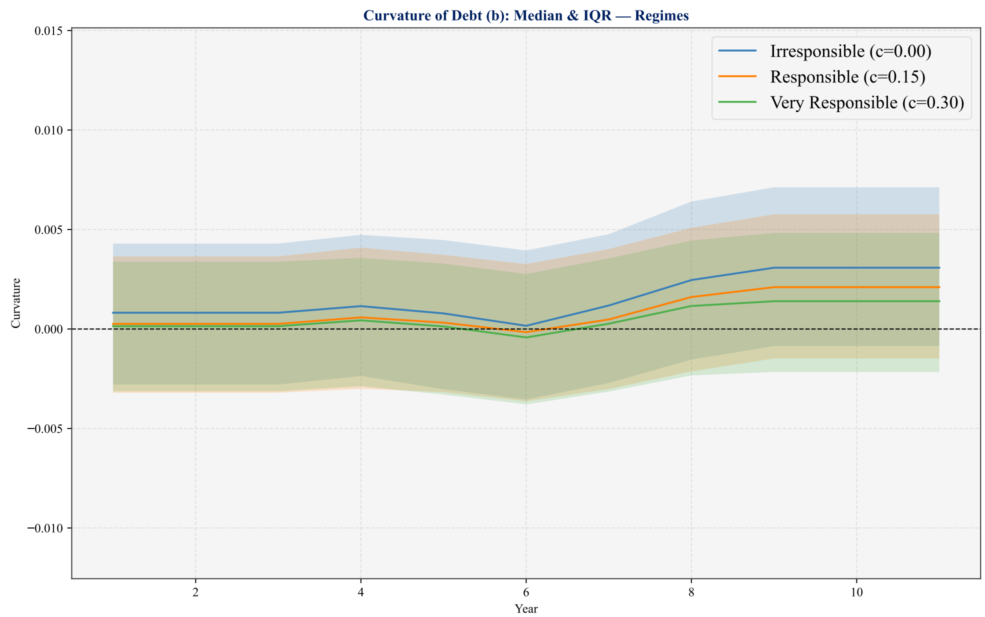


Figure 2: Curvature of Debt Path Projections

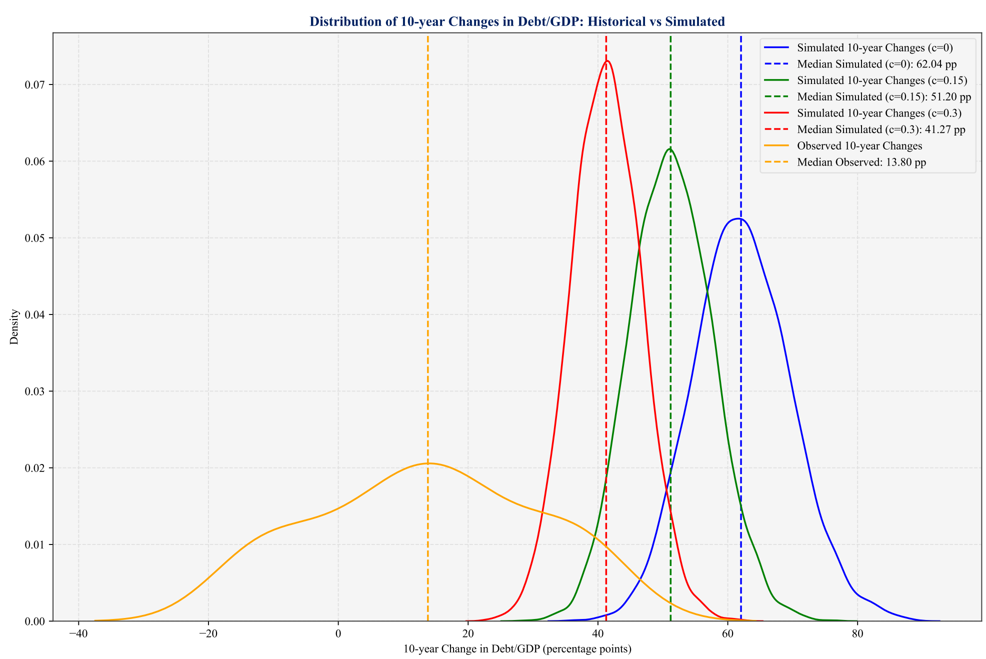


Figure 3: Distributions of 10-Year Changes in Debt, Historical and Projected

1.5 Addendum - Artificial Intelligence Boom

Some economists have suggested that AI technology, once integrated into the workforce and production, will lead to a sharp increase in productivity growth, and that this will boost g relative r, thereby staving off the unsustainable scenarios shown above. For example, Torsten Sløk, chief economist of Apollo, released an edition of his "Daily Spark" newsletter in July 2025 titled "AI Could Solve the US Fiscal Problem", based on CBO projections. The Penn Wharton Budget Model (PWBM) estimated (in a very preliminary analysis whose methodology has not been made public yet) that, through productivity and accordingly GDP gains, AI could reduce deficits by $400 billion (1.4% of 2024 GDP) between 2026 and 2035.[7].

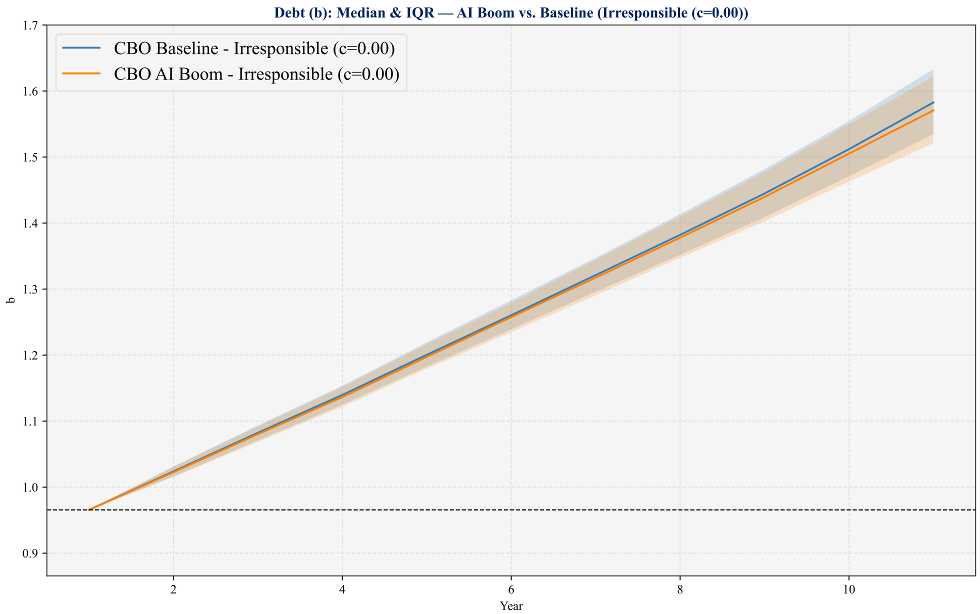
An attribute of our model is that we can simulate this scenario. In this case, we plug in CBO's projected growth rates under an AI boom, and again, generate a distribution of of debt paths.[8] We do that here; below we include two visual comparisons. First, in Subfigure 4a, we compare the future debt path distribution under $c = 0$ and 1) faster-than-baseline TFP growth vs. 2) baseline TFP growth. Second, in Subfigure **??**, we compare the debt path distribution under $c = 0.15$ and the alternative TFP growth scenarios.[9]

Our model's findings — which, we caution, does **only** consider the first order effects of an AI boom raising growth — suggest that AI, assuming significant productivity boosts, is not a cure-all for the debt sustainability challenges we face. Relative to CBO, in particular, we adopt post-OBBBA deficit projections, and the path of these deficit projections dominate in our model (post-OBBBA the projected deficit is significantly greater in magnitude than the pre-OBBBA CBO baseline projection). The gap in the fiscal outcome of the baseline vs. AI boom scenarios is also true to CBO's growth projections; the gap in our projected fiscal outcomes starts to widen as we approach 2035 (year 10 in the model), which is our operational horizon. The gap between CBO's projected growth and baseline is exponential (which makes sense, as pointed out by Larry Summers, "AI has the capacity to be self-improving", perhaps making it unique among technological advancements), suggesting that were we to extend our model outwards, we would likely see significant improvements to the fiscal situation under AI boom conditions as the horizon lengthens.

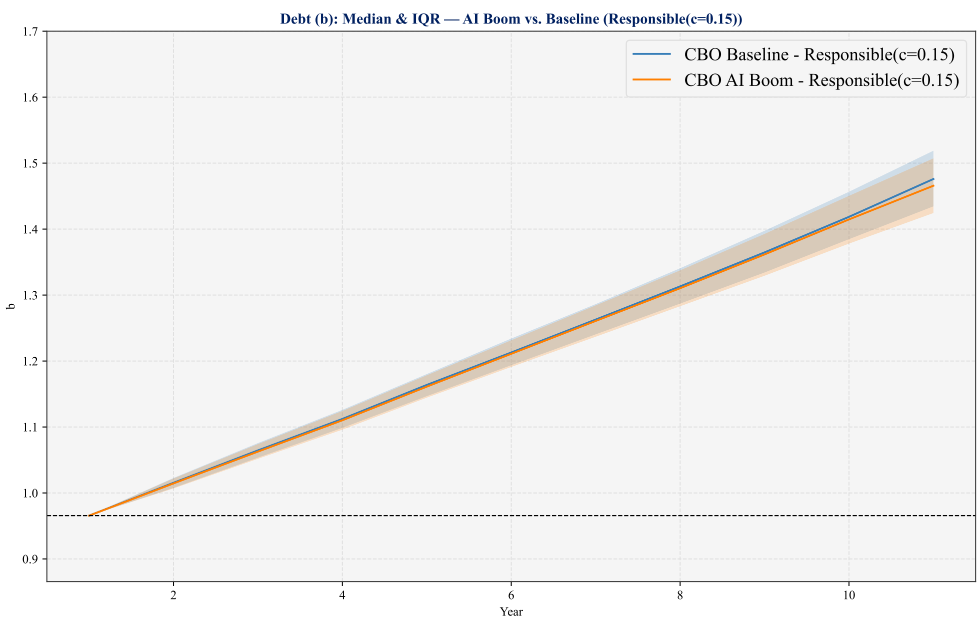
[7] Specifically, they find that AI will increase productivity and GDP by 1.5% by 2035, 3.5% by 2055, and 3.7% by 2075

[8] See Figure 6 for a plotted comparison of CBO's projected growth rates under an AI productivity boom and the baseline. To be precise, CBO reports real GDP on a *per capita basis*; to transform this number into a real GDP estimate, we multiply the forecasted real GDP per capita by the projected US population, as estimated by the US Census Bureau.

[9] We note that the Penn Wharton Budget Model forecasts AI boosts to productivity would also lower inflation; this would, in practice, also lower interest rates as the Federal Reserve would be less likely to raise the federal funds rate without elevated inflation to fight. A lower interest rate would make the debt path more depressed, suggesting that our estimates in the faster-than-baseline TFP growth scenario should be viewed as close to upper bounds for the debt-to-GDP ratio.



(a) Irresponsible Fiscal Regime ($c = 0$)



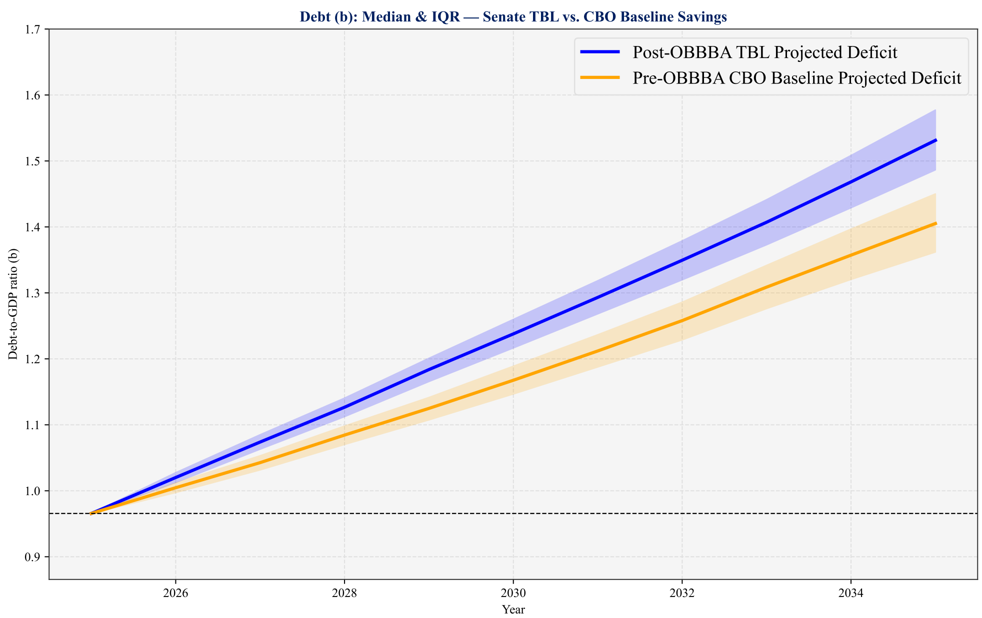
(b) Responsible Fiscal Regime ($c = 0.15$)

Figure 4: Debt Path Projections Under Alternative Growth Projections (AI Boom vs. Baseline)

1.6 Addendum - OBBBA Repeal

One of the clear advantages of our model that we have not explored is the ability to toggle different forecasts of s, in effect modeling the effects of deficit-financing forecasted policies. In the previous addendum, we illustrated how swapping out the CBO baseline real growth rate projections changes the results of our model, although such changes are muted because of the predominance of the projected mean deficit path. Here we adjust this vector, a_s, and see how our model reflects this changed input.

In this example, consider if we were in the position of a policymaker while Congress was deliberating over passage of the One Big Beautiful Bill Act (OBBBA). Our starting place for projecting out the model would be the CBO baseline published prior to the OBBBA passage. Swapping out these two mean deficit projections, holding all else constant with $c = 0.075$ (splitting the difference between a 'responsible' and 'irresponsible' Congress) we can produce the following debt paths, with median and interquartile range (IQR) plotted:



Projections like the above would allow policymakers to make probabilistic statements such as the following: "We do not forecast a debt path in the interquartile (between the 25th and 7th percentiles) of future distributions where the One Big Beautiful Bill Act (OBBBA), once permanently adopted, would not push our fiscal future into a worse position than a baseline fiscal situation, all else controlled."

2 Strengths and Weaknesses of our Model

Any model involves tradeoffs and the simplification of reality. In our model, which we view as a starting point, we have opted to emphasize first principles and model straightforward, easily-interpretable laws of motion for r, g, s, and b. This straightforwardness is a strength; we believe that this model is more easily digestible to policymakers than other complex (and also deeply helpful) macroeconomic models and, given that the goal of this exercise is to quantify debt sustainability in a way helpful for policymakers with direct control over our fiscal situation, we consider this a strength —- even if this simplification comes, in part, at the expense of model richness.

Additionally, the model gives us full control over the mean and distribution of the future path of our key macroeconomic variables r, s, and g. This means that we can use as the first moment of any of these variables projections from external sources; if a bill is proposed, and sources such as the CBO or Yale Budget Lab score these proposed policies on their fiscal/growth/interest rate effects, we can very easily construct counterfactual debt paths that allow us to compare across differing policy baselines. Importantly, this also allows us to model what would happen if, as an example, with the election of a new President in 2028, c went from 0 to 0.15 (i.e., Congress continues its fiscal recklessness until a new administration comes to power, and at that point it returns to its 1984-2003 approach to fiscal matters).

One weakness of our model, as a result of its simplification, is our reliance on linear, independent modeling of r, g, and s (although we do allow for some feedback between r and s). We know that the economy is deeply interconnected; changes in b will result in crowd out which will not only raise r but also, as a result of higher borrowing costs, lower g. Our model accounts (in a simplistic way) for this first step in crowd out; however, we would have to enrich our current framework to break out interest rate effects separately into 1) effects on the policy / federal funds rate and 2) private borrowing rates, each of which would presumably affect growth differently. Thus, while the (infamous) findings of Reinhart and Rogoff have been debunked, there is no consensus on the relationship between b and g — meaning it is unclear whether by excluding any causal effect of b on g we risk over- or under-estimating the severity of debt unsustainability (Rahman, Ismail, and Ridzuan 2019).

Furthermore, our model does not take a stance on whether the relationship between debt and interest rates (represented by the coefficient β_r) will fundamentally change in the future; this remains open speculation as we continue to hurdle into uncharted fiscal waters, but it is certainly possible that in instances of a debt spiral, the relationship between b and r is not linear but instead exponential.

One final weakness is that our model does not include the granularity to model changes in the composition of outstanding debt. For example, we make no distinction between the government issuing long-term treasuries or short-term treasures to fund itself; instead, we model a fixed and time-invariant parameter σ which captures the relative strength of the past average interest rate (intended to reflect the average interest rate across all outstanding debt) in determining the future average interest rate. This lack of enrichment prevents our *accurate* projection of debt servicing as a share of GDP (hence our reliance on the curvature of the debt path, which we believe captures in effect the same phenomenon).

What we would like to emphasize is that none of these weaknesses are fundamental to our model; instead, they require enrichments to the template we've laid down and we encourage alterations to this framework to incorporate more non-linear, interrelated, and complex relationships to further improve this model over time.

3 Appendix

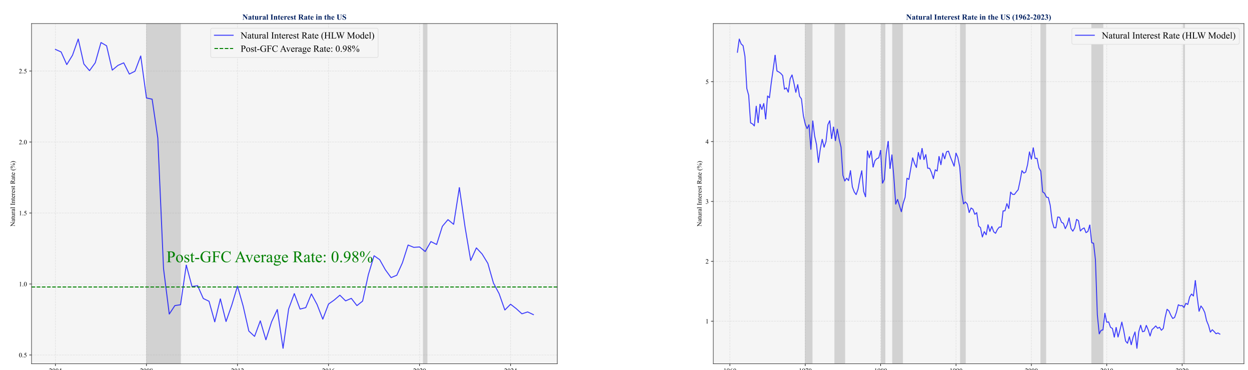
3.1 Previous Estimates of Debt's Effect on Interest Rates

Source	**Estimation Period**	**Effect of 1 pct. pt. increase in debt/GDP (basis points)**	**Methodology**	**Term Premium or Expected Short-Rate**
Engen and Hubbard (2004)	1976–2003	3 bps	OLS and VAR	both
Gale and Orszag (2004)	1976–2004	4–6 bps	OLS	both
Kinoshita (2006)	1971–2004, international data	4–5 bps	Panel regression	both
Laubach (2009)	1976–2006	3–4 bps	OLS	both
Chadha et al. (2013)	1986–2008	2 bps	OLS	both
Gamber and Seliski (2019)	1976–2017	2–3 bps	OLS	both
Coenen et al. (2012)	N/A	1 bp	Calibrated structural model	Expected short-rate (r^*)
Mian et al. (2022)	N/A	1–2 bps	Calibrated structural model	Expected short-rate (r^*)
Li and Wei (2013)*	1994–2007	6 bps	Estimated term structure model	Term premium

Table 3: Estimates of the effect of a 1 percentage point increase in debt-to-GDP on interest rates

3.2 Detailed Explanation of Modeling r

First, here we have plotted the current model r^* estimates from the Holston-Laubach-Williams (2023) model:



(a) Recent History of r^*

(b) Longer History of r^*

Figure 5

Second, to illustrate how ρ works, consider three simple examples: $\rho = 0$, $\rho = \frac{1}{2}$, and $\rho = 1$:

$$\rho = 0 \rightarrow r_t = \bar{r} + \beta_r b_{t-1} + e_t^r \tag{10}$$

$$\rho = \frac{1}{2} \rightarrow r_t = \frac{1}{2}\bar{r} + \frac{1}{2}r_{t-1} + \beta_r(b_{t-1} - \frac{1}{2}b_{t-2}) + e_t^r \tag{11}$$

$$\rho = 1 \rightarrow r_t = r_{t-1} + \beta_r(b_{t-1} - b_{t-2}) + e_t^r \tag{12}$$

Lastly, below I've included the regression results from estimating Equation 9, under three sample specifications: 1) entire sample, 2) pre-GFC, and 3) post-GFC:

Table 4: ACM 10y Term Premium: First Differences (HAC SEs)

	Full sample	Post-GFC	Pre-GFC
const	-0.008	-0.038	-0.005
	(0.021)	(0.037)	(0.025)
d_b_lag ($\beta_{\mathbf{r}}$)	**0.017***	**0.013**	**0.016**
	(0.009)	(0.010)	(0.047)
d_b_lag2	-0.001	0.001	0.001
	(0.007)	(0.006)	(0.038)
d_gap	-0.017	0.018	-0.031
	(0.021)	(0.015)	(0.034)
d_primary	0.015	0.019	0.018
	(0.011)	(0.012)	(0.039)
d_r_lag (ρ)	**0.158****	**0.151**	**0.152***
	(0.071)	(0.121)	(0.085)
R-squared	0.032	0.043	0.030
N	219	66	148

3.3 Detailed Explanation of Modeling s

To illustrate how c works in our model, we can evaluate the two extreme cases, either where $c = 0$ or where $c = 1$:

$$c = 0 \rightarrow s_t = (1 - c)a_s + c(r_t^{av} - g_t)b_{t-1} + e_t^s = a_s + e_t^s \tag{13}$$

$$c = 1 \rightarrow s_t = (1 - c)a_s + c(r_t^{av} - g_t)b_{t-1} + e_t^s = (r_t^{av} - g_t)b_{t-1} + e_t^s \tag{14}$$

We ultimately adopt three values for c: 0 (total irresponsibility, matching findings by Auerbach and Yagan 2025), 0.15 (a return to pre-2003 fiscal solitude), and 3) 0.3.

3.4 CBO-Modeled Growth Gains from AI Productivity Boom

Below in Figure 6 we compare the CBO-modeled AI productivity boom-induced growth rate vs. the baseline projected growth rate.

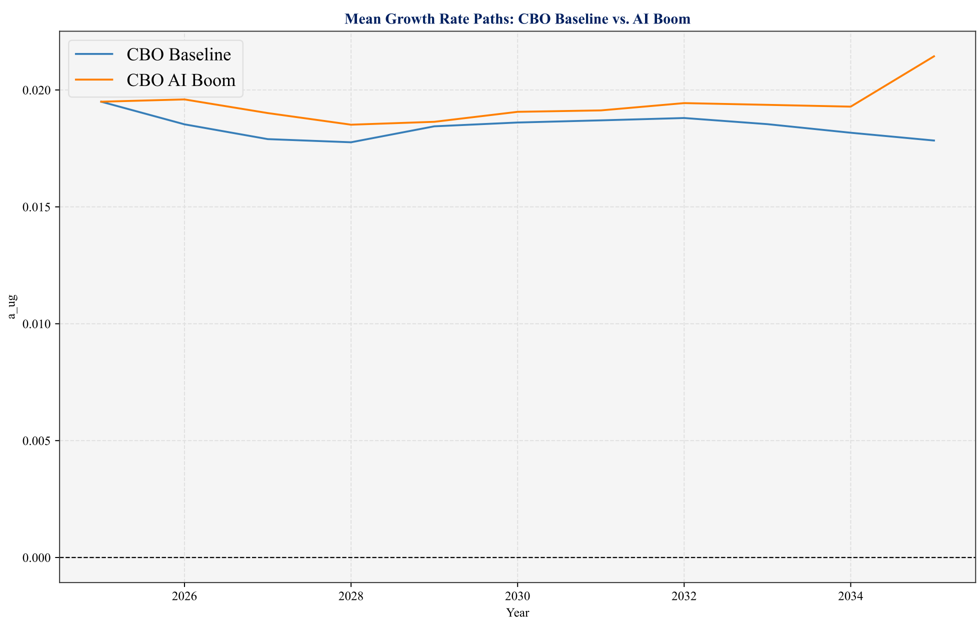


Figure 6: CBO AI Boom Growth Rate Path vs. CBO Basline Growth Rate